

Full classification of revivals on tori for separable coin

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1 Reduced Set of Revival Parameters

Observation 1 (Interchangeability of k and m). *The results are symmetric under the exchange $L_x \leftrightarrow L_y$: every parameter set $(\rho_x, \delta_x, \rho_y, \delta_y)$ that yields a revival of period T on $L_x \times L_y$ also yields a revival of the same period T on $L_y \times L_x$ with (ρ_x, δ_x) and (ρ_y, δ_y) interchanged. This follows from the symmetry of the separable coin $C = C_x \otimes C_y$ and the commutativity of the two shift operators. Consequently, only one orientation of each non-square torus needs to be tabulated; the $m \times k$ entries are obtained by swapping the x and y columns. The table below lists the reduced set.*

Table 1: Reduced exhaustive parameter sets yielding complete quantum revivals for separable quantum walks on $L_x \times L_y$ tori. Only one orientation of each non-square torus is listed; the $m \times k$ case follows by exchanging the x and y columns. Reflection parameters are exact fractions or surds; phase parameters are in units of π .

Torus	T	ρ_x	δ_x	ρ_y	δ_y
2×2	2	$1/2$	1	$1/2$	1
2×2	2	$1/2$	1	$1/2$	0
2×2	2	$1/2$	0	$1/2$	1
2×2	2	$1/2$	0	$1/2$	0
4×4	6	$3/4$	0	$3/4$	0
3×3	8	$2/3$	0	$2/3$	0
3×4	8	$2/3$	0	$1/2$	1
3×4	8	$2/3$	0	$1/2$	0
3×6	8	$2/3$	0	$2/3$	0

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Torus	T	ρ_x	δ_x	ρ_y	δ_y
4×4	8	$1/2$	1	$1/2$	1
4×4	8	$1/2$	1	$1/2$	0
4×4	8	$1/2$	0	$1/2$	1
4×4	8	$1/2$	0	$1/2$	0
4×6	8	$1/2$	1	$2/3$	0
4×6	8	$1/2$	0	$2/3$	0
6×6	8	$2/3$	0	$2/3$	0
3×3	10	$(3 - \sqrt{5})/4$	0	$(3 - \sqrt{5})/4$	0
3×6	10	$(3 - \sqrt{5})/4$	0	$(3 - \sqrt{5})/4$	0
6×6	10	$(3 - \sqrt{5})/4$	0	$(3 - \sqrt{5})/4$	0
3×3	12	$1/3$	$2/3$	$1/3$	$4/3$
3×3	12	$1/3$	$2/3$	$1/3$	$2/3$
3×3	12	$1/3$	$2/3$	$1/3$	0
3×3	12	$1/3$	0	$1/3$	$2/3$
3×3	12	$1/3$	0	$1/3$	0
3×3	12	$1/3$	$4/3$	$1/3$	$2/3$
3×3	12	$1/3$	$4/3$	$1/3$	0
3×3	12	$1/3$	0	$1/3$	$4/3$
3×3	12	$1/3$	$4/3$	$1/3$	$4/3$
3×4	12	$1/3$	0	$(2 - \sqrt{3})/4$	$1/2$
3×4	12	$1/3$	$4/3$	$(2 - \sqrt{3})/4$	$3/2$
3×4	12	$1/3$	$2/3$	$3/4$	1
3×4	12	$1/3$	$4/3$	$(2 - \sqrt{3})/4$	$1/2$
3×4	12	$1/3$	0	$3/4$	1
3×4	12	$1/3$	$2/3$	$(2 - \sqrt{3})/4$	$3/2$
3×4	12	$1/3$	$4/3$	$1/4$	1
3×4	12	$1/3$	$2/3$	$(2 - \sqrt{3})/4$	$1/2$
3×4	12	$1/3$	$4/3$	$1/4$	0
3×4	12	$1/3$	0	$(2 - \sqrt{3})/4$	$3/2$
3×4	12	$1/3$	$2/3$	$1/4$	1
3×4	12	$1/3$	$2/3$	$1/4$	0
3×4	12	$1/3$	0	$1/4$	1
3×4	12	$1/3$	0	$1/4$	0
3×4	12	$1/3$	$4/3$	$3/4$	1
3×6	12	$1/3$	0	$1/3$	$4/3$
3×6	12	$1/3$	0	$1/3$	$2/3$
3×6	12	$1/3$	$4/3$	$1/3$	$4/3$
3×6	12	$1/3$	$4/3$	$1/3$	$2/3$
3×6	12	$1/3$	$2/3$	$1/3$	$2/3$
3×6	12	$1/3$	$2/3$	$1/3$	0
3×6	12	$1/3$	0	$1/3$	0
3×6	12	$1/3$	$2/3$	$1/3$	$4/3$
3×6	12	$1/3$	$4/3$	$1/3$	0

Torus	T	ρ_x	δ_x	ρ_y	δ_y
4×4	12	$(2 - \sqrt{3})/4$	$3/2$	$1/4$	0
4×4	12	$(2 - \sqrt{3})/4$	$1/2$	$3/4$	1
4×4	12	$1/4$	0	$1/4$	1
4×4	12	$3/4$	1	$(2 - \sqrt{3})/4$	$1/2$
4×4	12	$1/4$	0	$1/4$	0
4×4	12	$1/4$	0	$(2 - \sqrt{3})/4$	$3/2$
4×4	12	$1/4$	1	$3/4$	1
4×4	12	$(2 - \sqrt{3})/4$	$1/2$	$(2 - \sqrt{3})/4$	$3/2$
4×4	12	$3/4$	1	$1/4$	1
4×4	12	$(2 - \sqrt{3})/4$	$1/2$	$(2 - \sqrt{3})/4$	$1/2$
4×4	12	$3/4$	1	$1/4$	0
4×4	12	$3/4$	1	$(2 - \sqrt{3})/4$	$3/2$
4×4	12	$1/4$	1	$(2 - \sqrt{3})/4$	$3/2$
4×4	12	$1/4$	1	$(2 - \sqrt{3})/4$	$1/2$
4×4	12	$(2 - \sqrt{3})/4$	$3/2$	$3/4$	1
4×4	12	$(2 - \sqrt{3})/4$	$1/2$	$1/4$	0
4×4	12	$1/4$	1	$1/4$	1
4×4	12	$1/4$	1	$1/4$	0
4×4	12	$1/4$	0	$3/4$	1
4×4	12	$(2 - \sqrt{3})/4$	$3/2$	$(2 - \sqrt{3})/4$	$3/2$
4×4	12	$(2 - \sqrt{3})/4$	$3/2$	$(2 - \sqrt{3})/4$	$1/2$
4×4	12	$(2 - \sqrt{3})/4$	$1/2$	$1/4$	1
4×4	12	$1/4$	0	$(2 - \sqrt{3})/4$	$1/2$
4×4	12	$(2 - \sqrt{3})/4$	$3/2$	$1/4$	1
4×4	12	$3/4$	1	$3/4$	1
4×6	12	$1/4$	0	$1/3$	$2/3$
4×6	12	$1/4$	0	$1/3$	0
4×6	12	$(2 - \sqrt{3})/4$	$3/2$	$1/3$	$4/3$
4×6	12	$(2 - \sqrt{3})/4$	$3/2$	$1/3$	$2/3$
4×6	12	$1/4$	0	$1/3$	$4/3$
4×6	12	$3/4$	1	$1/3$	0
4×6	12	$1/4$	1	$1/3$	$4/3$
4×6	12	$1/4$	1	$1/3$	$2/3$
4×6	12	$1/4$	1	$1/3$	0
4×6	12	$(2 - \sqrt{3})/4$	$1/2$	$1/3$	$4/3$
4×6	12	$(2 - \sqrt{3})/4$	$1/2$	$1/3$	$2/3$
4×6	12	$3/4$	1	$1/3$	$4/3$
4×6	12	$(2 - \sqrt{3})/4$	$1/2$	$1/3$	0
4×6	12	$3/4$	1	$1/3$	$2/3$
4×6	12	$(2 - \sqrt{3})/4$	$3/2$	$1/3$	0
6×6	12	$1/3$	$2/3$	$1/3$	$4/3$
6×6	12	$1/3$	$2/3$	$1/3$	$2/3$

Torus	T	ρ_x	δ_x	ρ_y	δ_y
6×6	12	$1/3$	$2/3$	$1/3$	0
6×6	12	$1/3$	0	$1/3$	0
6×6	12	$1/3$	$4/3$	$1/3$	0
6×6	12	$1/3$	0	$1/3$	$4/3$
6×6	12	$1/3$	0	$1/3$	$2/3$
6×6	12	$1/3$	$4/3$	$1/3$	$4/3$
6×6	12	$1/3$	$4/3$	$1/3$	$2/3$
3×3	16	$(2 - \sqrt{2})/4$	0	$(2 - \sqrt{2})/4$	0
3×4	16	$(2 - \sqrt{2})/4$	0	$(2 + \sqrt{2})/4$	0
3×4	16	$(2 - \sqrt{2})/4$	0	$(2 - \sqrt{2})/4$	0
3×4	16	$(2 - \sqrt{2})/4$	0	$(2 - \sqrt{2})/2$	$1/2$
3×6	16	$(2 - \sqrt{2})/4$	0	$(2 - \sqrt{2})/4$	0
4×4	16	$(2 - \sqrt{2})/4$	0	$(2 - \sqrt{2})/2$	$1/2$
4×4	16	$(2 + \sqrt{2})/4$	0	$(2 + \sqrt{2})/4$	0
4×4	16	$(2 - \sqrt{2})/2$	$1/2$	$(2 - \sqrt{2})/4$	0
4×4	16	$(2 - \sqrt{2})/4$	0	$(2 + \sqrt{2})/4$	0
4×4	16	$(2 - \sqrt{2})/2$	$1/2$	$(2 - \sqrt{2})/2$	$1/2$
4×4	16	$(2 + \sqrt{2})/4$	0	$(2 - \sqrt{2})/2$	$1/2$
4×4	16	$(2 + \sqrt{2})/4$	0	$(2 - \sqrt{2})/4$	0
4×4	16	$(2 - \sqrt{2})/2$	$1/2$	$(2 + \sqrt{2})/4$	0
4×4	16	$(2 - \sqrt{2})/4$	0	$(2 - \sqrt{2})/4$	0
4×6	16	$(2 - \sqrt{2})/2$	$1/2$	$(2 - \sqrt{2})/4$	0
4×6	16	$(2 - \sqrt{2})/4$	0	$(2 - \sqrt{2})/4$	0
4×6	16	$(2 + \sqrt{2})/4$	0	$(2 - \sqrt{2})/4$	0
6×6	16	$(2 - \sqrt{2})/4$	0	$(2 - \sqrt{2})/4$	0
3×3	20	$(5 - \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	0
3×4	20	$(5 - \sqrt{5})/8$	0	$(5 - \sqrt{5})/4$	0
3×4	20	$(5 - \sqrt{5})/8$	0	$(5 + \sqrt{5})/8$	1
3×4	20	$(5 - \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	0
4×4	20	$(5 - \sqrt{5})/4$	0	$(5 - \sqrt{5})/4$	0
4×4	20	$(5 + \sqrt{5})/8$	1	$(5 - \sqrt{5})/8$	0
4×4	20	$(5 - \sqrt{5})/8$	0	$(5 + \sqrt{5})/8$	1
4×4	20	$(5 - \sqrt{5})/8$	0	$(5 - \sqrt{5})/4$	0
4×4	20	$(5 + \sqrt{5})/8$	1	$(5 - \sqrt{5})/4$	0
4×4	20	$(5 - \sqrt{5})/4$	0	$(5 - \sqrt{5})/8$	0
4×4	20	$(5 + \sqrt{5})/8$	1	$(5 + \sqrt{5})/8$	1
4×4	20	$(5 - \sqrt{5})/4$	0	$(5 + \sqrt{5})/8$	1
4×4	20	$(5 - \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	0
3×3	24	$2/3$	$2/3$	$2/3$	$2/3$
3×3	24	$2/3$	$4/3$	$(5 - \sqrt{21})/10$	0
3×3	24	$2/3$	$2/3$	$2/3$	$4/3$
3×3	24	$2/3$	$2/3$	$(5 - \sqrt{21})/10$	0

Torus	T	ρ_x	δ_x	ρ_y	δ_y
3×3	24	$(5 - \sqrt{21})/10$	0	$2/3$	$4/3$
3×3	24	$(5 - \sqrt{21})/10$	0	$2/3$	$2/3$
3×3	24	$2/3$	$4/3$	$2/3$	$4/3$
3×3	24	$2/3$	$4/3$	$2/3$	$2/3$
3×3	24	$(5 - \sqrt{21})/10$	0	$(5 - \sqrt{21})/10$	0
3×4	24	$2/3$	$4/3$	$(3 + \sqrt{3})/6$	0
3×4	24	$2/3$	$4/3$	$1/2$	$1/2$
3×4	24	$(5 - \sqrt{21})/10$	0	$(3 + \sqrt{3})/6$	0
3×4	24	$2/3$	$4/3$	$(3 - \sqrt{3})/6$	0
3×4	24	$2/3$	$2/3$	$1/2$	$3/2$
3×4	24	$2/3$	$2/3$	$1/2$	$1/2$
3×4	24	$2/3$	$2/3$	$(3 + \sqrt{3})/6$	0
3×4	24	$(5 - \sqrt{21})/10$	0	$(3 - \sqrt{3})/6$	0
3×4	24	$(5 - \sqrt{21})/10$	0	$1/2$	$3/2$
3×4	24	$(5 - \sqrt{21})/10$	0	$1/2$	$1/2$
3×4	24	$2/3$	$2/3$	$(3 - \sqrt{3})/6$	0
3×4	24	$2/3$	$4/3$	$1/2$	$3/2$
3×8	24	$(5 - \sqrt{21})/10$	0	$1/2$	$1/2$
3×8	24	$2/3$	$2/3$	$1/2$	1
3×8	24	$2/3$	$4/3$	$1/2$	$3/2$
3×8	24	$2/3$	$4/3$	$1/2$	$1/2$
3×8	24	$(5 - \sqrt{21})/10$	0	$1/2$	$3/2$
3×8	24	$(5 - \sqrt{21})/10$	0	$1/2$	0
3×8	24	$2/3$	$2/3$	$1/2$	$1/2$
3×8	24	$2/3$	$4/3$	$1/2$	1
3×8	24	$2/3$	$4/3$	$1/2$	0
3×8	24	$(5 - \sqrt{21})/10$	0	$1/2$	1
3×8	24	$2/3$	$2/3$	$1/2$	$3/2$
3×8	24	$2/3$	$2/3$	$1/2$	0
4×4	24	$1/2$	$1/2$	$1/2$	$1/2$
4×4	24	$1/2$	$1/2$	$1/2$	$3/2$
4×4	24	$1/2$	$1/2$	$(3 + \sqrt{3})/6$	0
4×4	24	$1/2$	$3/2$	$(3 - \sqrt{3})/6$	0
4×4	24	$1/2$	$1/2$	$(3 - \sqrt{3})/6$	0
4×4	24	$1/2$	$3/2$	$1/2$	$3/2$
4×4	24	$1/2$	$3/2$	$1/2$	$1/2$
4×4	24	$1/2$	0	$(3 + \sqrt{3})/6$	0
4×4	24	$1/2$	0	$(3 - \sqrt{3})/6$	0
4×4	24	$1/2$	$3/2$	$(3 + \sqrt{3})/6$	0
4×4	24	$1/2$	0	$1/2$	$1/2$
4×4	24	$1/2$	0	$1/2$	$3/2$
4×4	24	$1/2$	$3/2$	$(3 - \sqrt{3})/6$	0

Torus	T	ρ_x	δ_x	ρ_y	δ_y
4×4	24	$1/2$	0	$1/2$	0
4×4	24	$1/2$	$3/2$	$1/2$	0
4×4	24	$1/2$	$1/2$	$1/2$	0
4×8	24	$(3 + \sqrt{3})/6$	0	$1/2$	1
4×8	24	$1/2$	$1/2$	$1/2$	1
4×8	24	$1/2$	$1/2$	$1/2$	0
4×8	24	$1/2$	$3/2$	$1/2$	$3/2$
4×8	24	$1/2$	$3/2$	$1/2$	$1/2$
4×8	24	$(3 + \sqrt{3})/6$	0	$1/2$	$1/2$
4×8	24	$(3 - \sqrt{3})/6$	0	$1/2$	$3/2$
4×8	24	$(3 - \sqrt{3})/6$	0	$1/2$	$1/2$
4×8	24	$1/2$	$3/2$	$1/2$	1
4×8	24	$1/2$	$3/2$	$1/2$	0
4×8	24	$(3 + \sqrt{3})/6$	0	$1/2$	$3/2$
4×8	24	$(3 + \sqrt{3})/6$	0	$1/2$	0
4×8	24	$1/2$	$1/2$	$1/2$	$3/2$
4×8	24	$(3 - \sqrt{3})/6$	0	$1/2$	1
4×8	24	$1/2$	$1/2$	$1/2$	$1/2$
4×8	24	$(3 - \sqrt{3})/6$	0	$1/2$	0
8×8	24	$1/2$	$1/2$	$1/2$	1
8×8	24	$1/2$	0	$1/2$	$3/2$
8×8	24	$1/2$	0	$1/2$	0
8×8	24	$1/2$	$3/2$	$1/2$	1
8×8	24	$1/2$	$3/2$	$1/2$	0
8×8	24	$1/2$	1	$1/2$	$3/2$
8×8	24	$1/2$	1	$1/2$	$1/2$
8×8	24	$1/2$	0	$1/2$	1
8×8	24	$1/2$	$1/2$	$1/2$	$1/2$
8×8	24	$1/2$	1	$1/2$	1
8×8	24	$1/2$	1	$1/2$	0
8×8	24	$1/2$	$1/2$	$1/2$	$3/2$
8×8	24	$1/2$	$1/2$	$1/2$	0
8×8	24	$1/2$	0	$1/2$	$1/2$
8×8	24	$1/2$	$3/2$	$1/2$	$3/2$
8×8	24	$1/2$	$3/2$	$1/2$	$1/2$
3×3	30	0.460655	$2/3$	0.460655	$4/3$
3×3	30	0.460655	$4/3$	0.460655	$4/3$
3×3	30	0.460655	$4/3$	0.460655	$2/3$
3×3	30	0.460655	$2/3$	0.460655	$2/3$
5×5	60	$(5 - \sqrt{5})/8$	$4/5$	$(5 - \sqrt{5})/8$	$4/5$
5×5	60	$(3 + \sqrt{5})/8$	$4/5$	$(5 - \sqrt{5})/8$	$2/5$
5×5	60	$(3 + \sqrt{5})/8$	$2/5$	$(3 + \sqrt{5})/8$	0
5×5	60	$(3 + \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	0

Torus	T	ρ_x	δ_x	ρ_y	δ_y
5×5	60	$(5 - \sqrt{5})/8$	$4/5$	$(5 - \sqrt{5})/8$	0
5×5	60	$(3 + \sqrt{5})/8$	$2/5$	$(5 - \sqrt{5})/8$	$4/5$
5×5	60	$(3 + \sqrt{5})/8$	$2/5$	$(3 + \sqrt{5})/8$	$4/5$
5×5	60	$(5 - \sqrt{5})/8$	$2/5$	$(3 + \sqrt{5})/8$	$2/5$
5×5	60	$(3 + \sqrt{5})/8$	$4/5$	$(3 + \sqrt{5})/8$	$4/5$
5×5	60	$(5 - \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	$2/5$
5×5	60	$(3 + \sqrt{5})/8$	$4/5$	$(5 - \sqrt{5})/8$	$4/5$
5×5	60	$(3 + \sqrt{5})/8$	$2/5$	$(5 - \sqrt{5})/8$	0
5×5	60	$(3 + \sqrt{5})/8$	0	$(3 + \sqrt{5})/8$	$2/5$
5×5	60	$(3 + \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	$2/5$
5×5	60	$(5 - \sqrt{5})/8$	$4/5$	$(3 + \sqrt{5})/8$	$2/5$
5×5	60	$(3 + \sqrt{5})/8$	$4/5$	$(3 + \sqrt{5})/8$	0
5×5	60	$(5 - \sqrt{5})/8$	$4/5$	$(5 - \sqrt{5})/8$	$2/5$
5×5	60	$(3 + \sqrt{5})/8$	$4/5$	$(5 - \sqrt{5})/8$	0
5×5	60	$(5 - \sqrt{5})/8$	0	$(3 + \sqrt{5})/8$	$4/5$
5×5	60	$(5 - \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	$4/5$
5×5	60	$(5 - \sqrt{5})/8$	$2/5$	$(5 - \sqrt{5})/8$	$2/5$
5×5	60	$(3 + \sqrt{5})/8$	$2/5$	$(5 - \sqrt{5})/8$	$2/5$
5×5	60	$(3 + \sqrt{5})/8$	$2/5$	$(3 + \sqrt{5})/8$	$2/5$
5×5	60	$(5 - \sqrt{5})/8$	0	$(3 + \sqrt{5})/8$	0
5×5	60	$(3 + \sqrt{5})/8$	$4/5$	$(3 + \sqrt{5})/8$	$2/5$
5×5	60	$(5 - \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	0
5×5	60	$(3 + \sqrt{5})/8$	0	$(3 + \sqrt{5})/8$	0
5×5	60	$(5 - \sqrt{5})/8$	$4/5$	$(3 + \sqrt{5})/8$	0
5×5	60	$(5 - \sqrt{5})/8$	$2/5$	$(3 + \sqrt{5})/8$	$4/5$
5×5	60	$(5 - \sqrt{5})/8$	$2/5$	$(5 - \sqrt{5})/8$	$4/5$
5×5	60	$(5 - \sqrt{5})/8$	0	$(3 + \sqrt{5})/8$	$2/5$
5×5	60	$(5 - \sqrt{5})/8$	$2/5$	$(3 + \sqrt{5})/8$	0
5×5	60	$(3 + \sqrt{5})/8$	0	$(3 + \sqrt{5})/8$	$4/5$
5×5	60	$(5 - \sqrt{5})/8$	$2/5$	$(5 - \sqrt{5})/8$	0
5×5	60	$(3 + \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	$4/5$
5×5	60	$(5 - \sqrt{5})/8$	$4/5$	$(3 + \sqrt{5})/8$	$4/5$
5×10	60	$(5 - \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	0
5×10	60	$(5 - \sqrt{5})/8$	$4/5$	$(3 + \sqrt{5})/8$	0
5×10	60	$(5 - \sqrt{5})/8$	0	$(3 + \sqrt{5})/8$	0
5×10	60	$(5 - \sqrt{5})/8$	$2/5$	$(5 - \sqrt{5})/8$	0
5×10	60	$(3 + \sqrt{5})/8$	$2/5$	$(3 + \sqrt{5})/8$	0
5×10	60	$(3 + \sqrt{5})/8$	$2/5$	$(5 - \sqrt{5})/8$	0
5×10	60	$(3 + \sqrt{5})/8$	$4/5$	$(3 + \sqrt{5})/8$	0
5×10	60	$(3 + \sqrt{5})/8$	0	$(3 + \sqrt{5})/8$	0
5×10	60	$(3 + \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	0

Torus	T	ρ_x	δ_x	ρ_y	δ_y
5×10	60	$(3 + \sqrt{5})/8$	$4/5$	$(5 - \sqrt{5})/8$	0
5×10	60	$(5 - \sqrt{5})/8$	$2/5$	$(3 + \sqrt{5})/8$	0
5×10	60	$(5 - \sqrt{5})/8$	$4/5$	$(5 - \sqrt{5})/8$	0
10×10	60	$(5 - \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	0
10×10	60	$(5 - \sqrt{5})/8$	0	$(3 + \sqrt{5})/8$	0
10×10	60	$(3 + \sqrt{5})/8$	0	$(3 + \sqrt{5})/8$	0
10×10	60	$(3 + \sqrt{5})/8$	0	$(5 - \sqrt{5})/8$	0